

a quantum computer. Quantum neural networks (QNNs) are used to model quantum computational models. A powerful tool for developing quantum computational models is TensorFlow Quantum, a library for hybrid quantum-classical machine learning. The presence of this and many other tools indicates that more and more quantum computing based AI solutions will be available in the near future.

Introduction to seaborn

Now let us learn to use seaborn. Seaborn is a Python data visualisation library based on Matplotlib. It provides functions for drawing visually appealing statistical images. Seaborn can be installed easily by using Anaconda Navigator, as explained in the previous article in this series. I have used the batting records in T20 international cricket from the website *ESPNcricinfo* to create a CSV (comma-separated values) file called *T20.csv*, which contains the following 15 columns: Player name, Career span, Matches, Innings, Not out, Total runs, Highest score, Average, Balls faced, Strike rate, Centuries, Fifties, Ducks, Number of 4s scored, and Number of 6s scored. We will use the library called Pandas to read this CSV file. Remember that we have already familiarised ourselves with Pandas. Now, consider the program shown in Figure 1.

Let us try to understand the working of the program. Lines 1 and 2 import the packages Pandas and seaborn. Line 3 reads the CSV file called *T20.csv* from the working directory of JupyterLab. Line 4 prints the metadata and the first line of data. Figure 1 shows this line, which displays the batting record of Virat Kohli, the highest run scorer in T20 international cricket matches. Line 5 stores the metadata and the first five rows of data in the CSV file *T20.csv* to another CSV file called *Best5.csv*. On execution of the code, you will see this file created in the working directory of JupyterLab. Line 6 sorts the CSV file in ascending

```
1 import pandas as pd
2 import seaborn as sns
3 df = pd.read_csv('./T20.csv')
4 print(df.head(1))
5 df.head(5).to_csv('Best5.csv')
6 df.sort_values("100", ascending=False).head(10).to_csv('Highest100.csv')
7 sns.scatterplot(data=df.iloc[:, [5, 7]].head(50), x='Ave', y='Runs')
```

	Player	Span	Mat	Inns	NO	Runs	HS	Ave	BF	SR
0	V Kohli (INDIA)	2010-2022	114	106	31	3958	122*	52.77	2865	138.15

	100	50	0	4s	6s
0	1	36	4	352	116

Figure 1: A simple example for seaborn

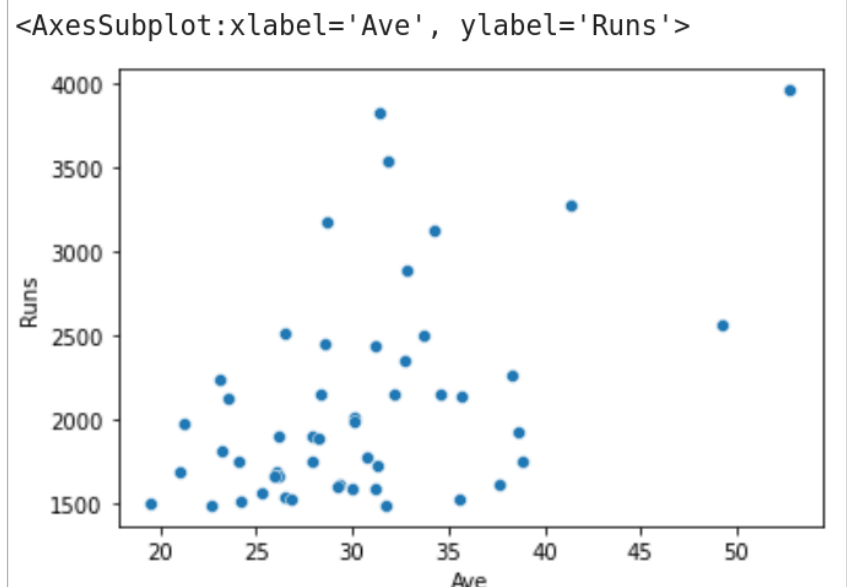


Figure 2: A scatter plot with seaborn

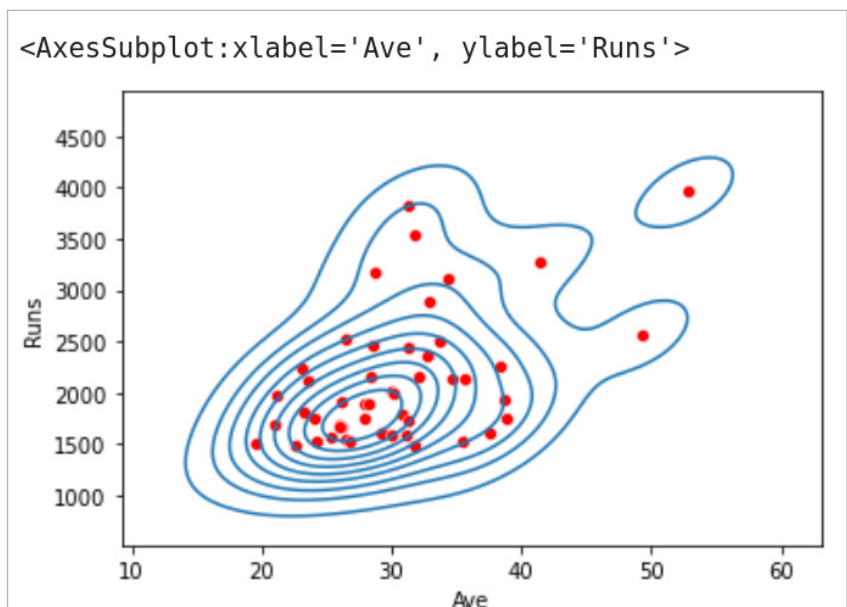


Figure 3: A KDE plot with seaborn